1.1 Classical Mechanics Models

Many of the models used for studying molecules with large numbers of atoms are said to be classical because they rely on assumptions that allow the use of Newton's equations for the motion of the nuclei. Coming to this subject for the first time, one may wonder why we don't use quantum mechanics, since that is a more modern and comprehensive theory of the behavior of matter. A system of N atoms can be described in a probabilistic way using the Schrödinger equation. For illustration, consider a water molecule which has 10 electrons and 3 nuclei, each of which is identified with a point in three dimensions, thus the basic object describing the probability distribution for an isolated water molecule is a complex-valued function of 39 variables. Denote a point in this space by the coordinates $q_{1,x}, q_{1,y}, q_{1,z}, q_{2,x}, q_{2,y}, q_{2,z}, \ldots, q_{13,x}, q_{13,y}, q_{13,z}$. The Schrödinger equation itself is a partial differential equation of the following form:

$$i\hbar \frac{\partial \Phi}{\partial t} = -\hbar^2 \sum_{j=1}^{13} \frac{1}{2\mu_j} \left(\frac{\partial^2 \Phi}{\partial q_{j,x}^2} + \frac{\partial^2 \Phi}{\partial q_{j,y}^2} + \frac{\partial^2 \Phi}{\partial q_{j,z}^2} \right) + U_{\mathrm{P}}(q_{1,x}, q_{1,y}, \ldots, q_{13,z}) \Phi. \tag{1.1}$$

i is the square root of minus 1, $\hbar$ is Planck's constant, μ_j is the mass of the jth particle, Φ is the unknown "wave-function," and U_{P} is the primitive atomic potential energy function which includes formulas that model the repulsion of electrons due to their similar charge and the attraction of electrons to the positively charged nucleus of each atom. We cannot write down the solution of Schrödinger's equation in a simple form, so we would need eventually to solve the equation numerically, using a computer. Ultimately, the problem with the quantum mechanical approach is one of computational complexity. It is impossible (the "curse of dimensionality") to handle the numerical discretization of a partial differential equation in many variables, due to the rapid explosion in the number of grid points or basis functions that would be needed for approximation. Even the water molecule is beyond the reach of a full scale quantum mechanical treatment. In quantum mechanics we know the fundamental laws describing the system quite well, but the large number of independent spatial dimensions rules out any straightforward attack on the problem. In the words of Paul Dirac, "The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known, and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble" [106].

An assumption that is often made in order to reduce the complexity of the problem is that the relatively light electrons of the system move instantaneously, entirely slaved to the motion of the heavier nuclei (this is called the *Born-Oppenheimer approximation*). It is then found under certain circumstances that the nuclear coordinates can be viewed as the approximate solutions of a system of ordinary differential equations involving a Born-Oppenheimer potential energy